

HIV-prEP CORE Anonymization Algorithms



Basic anonymization

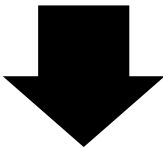
k-Anonymity Score, Re-identification Risk, l-Diversity, t-Closeness, Variance Retention, Information Loss (%), Processing Time, Memory Usage	K-Anonymization
Re-identification Risk, Variance Retention, Processing Time, Memory Usage	Pseudonymization
Re-identification Risk, Variance Retention, Processing Time, Memory Usage	Random Shuffling
Re-identification Risk, Variance Retention, Information Loss (%), Information Loss (%), Processing Time, Memory Usage	Generalization
Re-identification Risk, Variance Retention, Information Loss (%), Processing Time, Memory Usage	Supression
Re-identification Risk, Variance Retention, Processing Time, Memory Usage	Secondary Pseudonym Pools
Re-identification Risk, Variance Retention, Processing Time, Memory Usage	Column Segmentation



Fundamental Anonymization Checklist

Intermediate anonymization

Re-identification Risk, Variance Retention, Processing Time, Memory Usage	Top - Bottom Coding
Re-identification Risk, l-Diversity, Variance Retention, Correlation Analysis, Processing Time, Memory Usage	Micro-aggregation
Re-identification Risk, Variance Retention, Correlation Analysis, Processing Time, Memory Usage	Bucketing(Binning)
Re-identification Risk, Variance Retention, Processing Time,Memory Usage	Masking and Reduction
Re-identification Risk, Mean Absolute Error (MAE), Variance Retention, Processing Time, Memory Usage	Noise Addition (Data peturbation)
Re-identification Risk, Variance Retention, Processing Time, Memory Usage	Data Swapping



Statistical and Petrurbation Checklist

Advanced anonymization

Re-identification Risk, t-Closeness, Differential Privacy Level (?-DP), Variance Retention, Processing Time,Memory Usage	Differential Privacy
Re-identification Risk, Encryption Strength, Variance Retention, Processing Time, Memory Usage, Encryption Overhead	Homomorphic Encryption



Privacy-Preserving Computational-Based Checklist